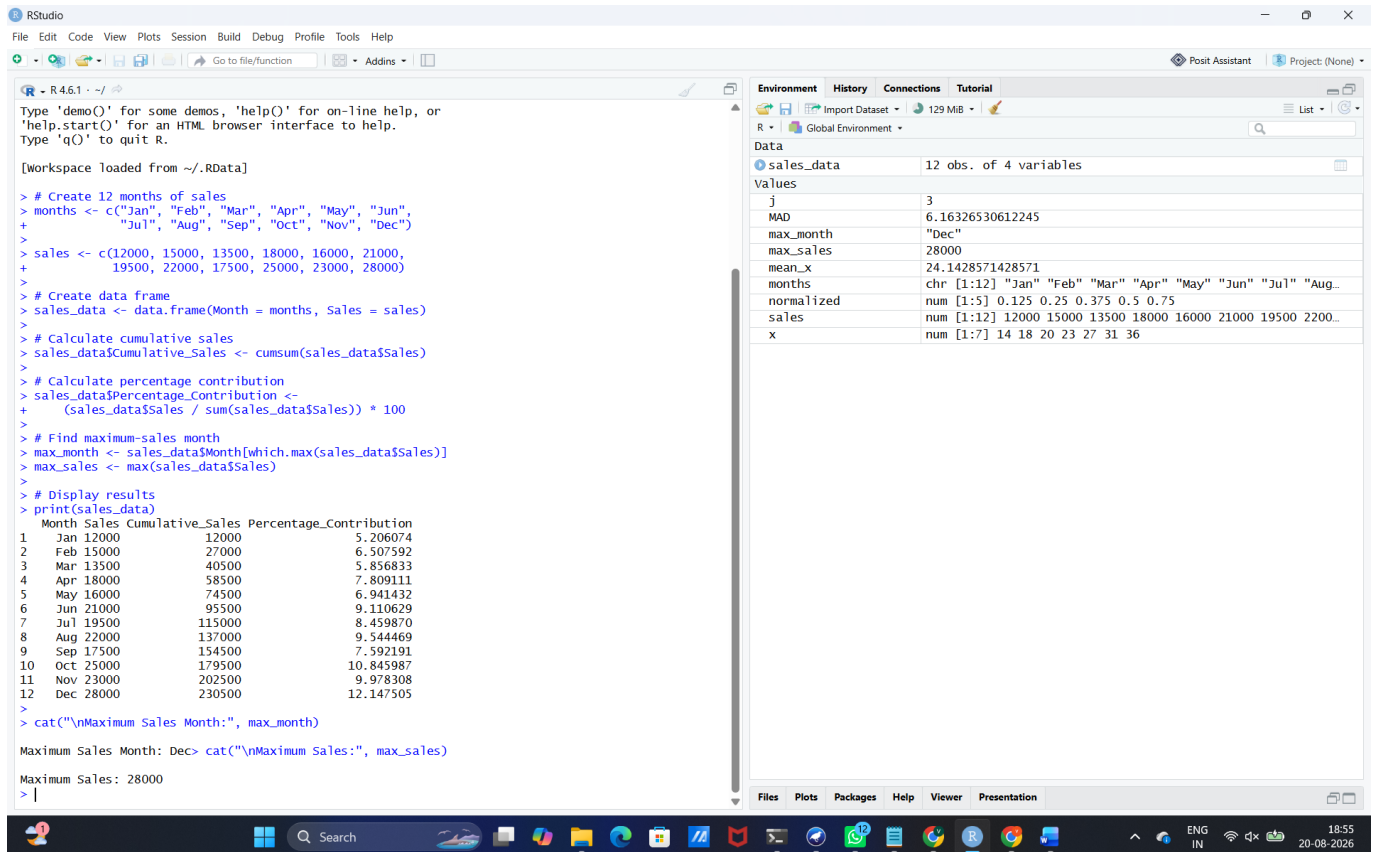


SET 5

1. R: Create 12 months of sales; calculate cumulative sales, percentage contribution and maximum-sales month.



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code to create 12 months of sales data, calculate cumulative sales, percentage contribution, and identify the maximum sales month. The code is as follows:

```
> # Create 12 months of sales
> months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
+            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
> sales <- c(12000, 15000, 13500, 18000, 16000, 21000,
+           19500, 22000, 17500, 25000, 23000, 28000)
> # Create data frame
> sales_data <- data.frame(Month = months, Sales = sales)
> # Calculate cumulative sales
> sales_data$Cumulative_Sales <- cumsum(sales_data$Sales)
> # Calculate percentage contribution
> sales_data$Percentage_Contribution <-
+   (sales_data$Sales / sum(sales_data$Sales)) * 100
> # Find maximum-sales month
> max_month <- sales_data$Month[which.max(sales_data$Sales)]
> max_sales <- max(sales_data$Sales)
> # Display results
> print(sales_data)
  Month Sales Cumulative_Sales Percentage_Contribution
1  Jan 12000          12000          5.206074
2  Feb 15000          27000          6.507592
3  Mar 13500          40500          5.856833
4  Apr 18000          58500          7.809111
5  May 16000          74500          6.941432
6  Jun 21000          95500          9.110629
7  Jul 19500         115000          8.459870
8  Aug 22000         137000          9.544469
9  Sep 17500         154500          7.592191
10 Oct 25000         179500         10.845987
11 Nov 23000         202500          9.978308
12 Dec 28000         230500         12.147505
>
> cat("\nMaximum Sales Month:", max_month)
Maximum Sales Month: Dec> cat("\nMaximum Sales:", max_sales)
Maximum Sales: 28000
>
```
- Environment:** Shows the 'sales_data' object with 12 observations and 4 variables. The variables and their values are:

Variable	Value
j	3
MAD	6.16326530612245
max_month	"Dec"
max_sales	28000
mean_x	24.1428571428571
months	chr [1:12] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug..."
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
sales	num [1:12] 12000 15000 13500 18000 16000 21000 19500 2200...
x	num [1:7] 14 18 20 23 27 31 36

2. R: Find the five-number summary for 15 observations and identify outliers using $1.5 \times \text{IQR}$.

The screenshot shows the RStudio interface with the following R code in the console:

```

[Workspace loaded from ~/.RData]
> # 15 observations
> data <- c(10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 27, 30, 32, 50)
>
> # Five-number summary
> summary(data)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  10.0   15.5   21.0   22.4   26.0   50.0
>
> # Calculate Q1, Median and Q3
> Q1 <- quantile(data, 0.25)
> Median <- median(data)
> Q3 <- quantile(data, 0.75)
>
> # Calculate IQR
> IQR_value <- IQR(data)
>
> # Calculate outlier limits
> lower_limit <- Q1 - 1.5 * IQR_value
> upper_limit <- Q3 + 1.5 * IQR_value
>
> # Identify outliers
> outliers <- data[data < lower_limit | data > upper_limit]
>
> # Display results
> cat("Minimum:", min(data), "\n")
Minimum: 10
> cat("Q1:", Q1, "\n")
Q1: 15.5
> cat("Median:", Median, "\n")
Median: 21
> cat("Q3:", Q3, "\n")
Q3: 26
> cat("Maximum:", max(data), "\n")
Maximum: 50
>
> cat("IQR:", IQR_value, "\n")
IQR: 10.5
> cat("Lower Limit:", lower_limit, "\n")
Lower Limit: -0.25
> cat("Upper Limit:", upper_limit, "\n")
Upper Limit: 41.75
> cat("Outliers:", outliers, "\n")
Outliers: 50

```

The Environment pane on the right shows the following variables:

Variable	Class	Value
data	num [1:15]	10 12 14 15 16 18 20 21 22 24 ...
IQR_value	num	10.5
j	num	3
lower_limit	Named num	-0.25
MAD	num	6.16326530612245
mean_x	num	24.1428571428571
Median	num	21
normalized	num [1:5]	0.125 0.25 0.375 0.5 0.75
outliers	num	50
Q1	Named num	15.5
Q3	Named num	26
upper_limit	Named num	41.8
x	num [1:7]	14 18 20 23 27 31 36

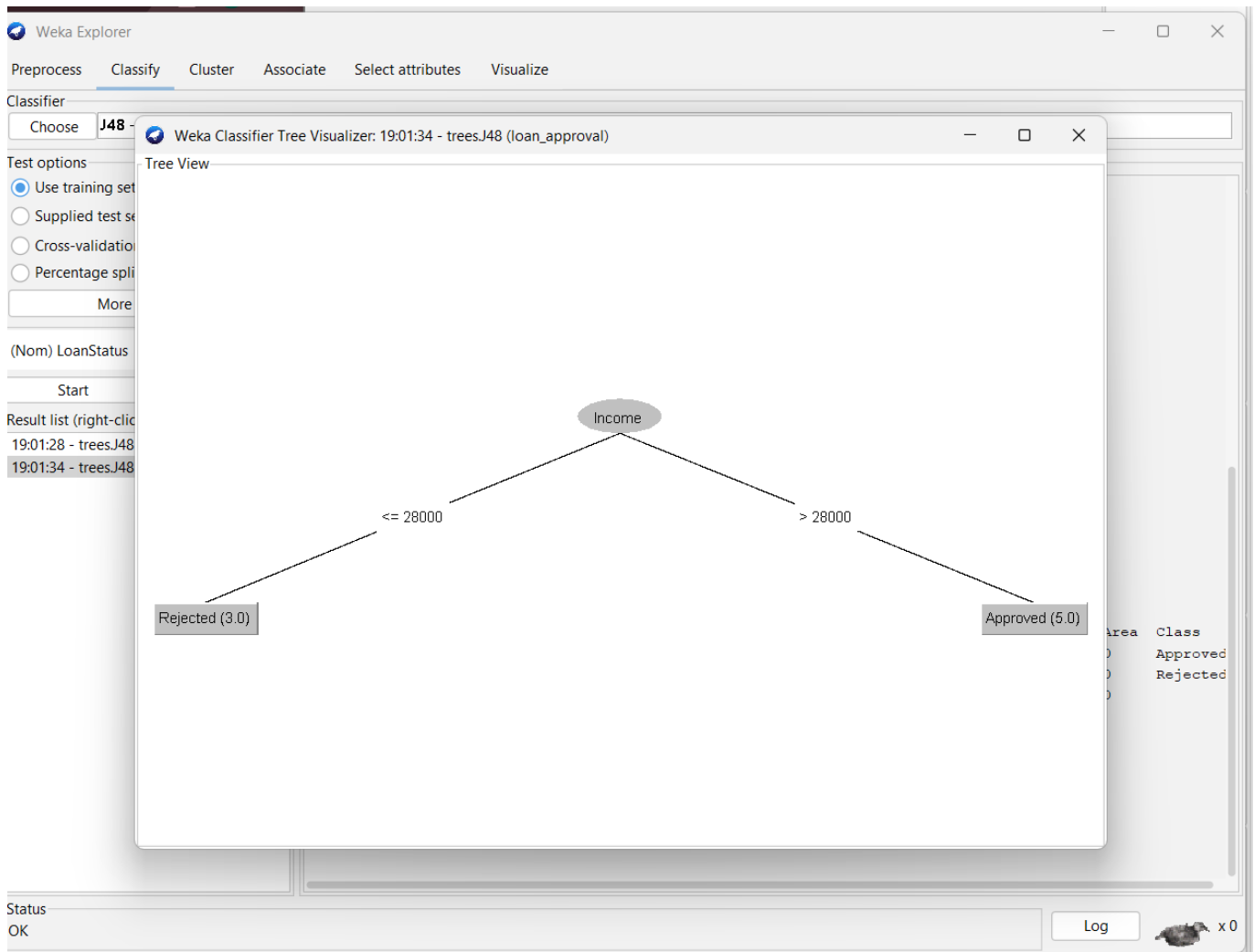
3. WEKA: Create loan-applicant ARFF data (Income, Age, Employment, LoanStatus); apply J48 and visualize.

Dataset: Loan Approval

```

@relation loan_approval
@attribute Income numeric
@attribute Age numeric
@attribute Employment {Employed,SelfEmployed,Unemployed}
@attribute LoanStatus {Approved,Rejected}
@data
25000,23,Employed,Rejected
32000,28,Employed,Approved
45000,35,Employed,Approved
28000,31,SelfEmployed,Rejected
60000,42,SelfEmployed,Approved
22000,25,Unemployed,Rejected
52000,38,Employed,Approved
35000,29,SelfEmployed,Approved

```



4. WEKA: Apply SMO to the loan dataset and compare accuracy and confusion matrix with J48.

Dataset: Loan Approval – SMO

Use the same Loan Approval ARFF dataset from Set 5 Q3.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) LoanStatus

Start

Stop

Result list (right-click for options)

19:03:54 - functions.SMO

Classifier output

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	6	75	%
Incorrectly Classified Instances	2	25	%
Kappa statistic	0.3846		
Mean absolute error	0.25		
Root mean squared error	0.5		
Relative absolute error	52.6316	%	
Root relative squared error	103.1421	%	
Total Number of Instances	8		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.667	0.714	1.000	0.833	0.488	0.667	0.714	Approved
	0.333	0.000	1.000	0.333	0.500	0.488	0.667	0.583	Rejected
Weighted Avg.	0.750	0.417	0.821	0.750	0.708	0.488	0.667	0.665	

=== Confusion Matrix ===

a b

<-- classified as

5 0 | a = Approved

2 1 | b = Rejected

Status

OK

Log

x 0